

Comparing Dry Etching and Wet Etching

Part A: Model Paragraph

Read the paragraph below and answer the questions.

Dry etching and wet etching are two fundamental techniques used in semiconductor fabrication, and they can be compared in terms of cost, precision, and equipment complexity. In wet etching, liquid chemical solutions are used, and the process is generally considered less expensive because simpler equipment, such as chemical baths, is required; in contrast, dry etching is performed using ionized gases in a vacuum environment, and more sophisticated and costly machinery, such as plasma systems, is needed. In terms of precision, wet etching is typically isotropic, meaning that material is removed uniformly in all directions and less control is achieved, whereas dry etching is highly anisotropic, so greater accuracy and finer feature definition are obtained. Additionally, while lower surface damage is usually observed in wet etching due to the absence of high-energy ions, a higher risk of damage is often associated with dry etching because ion bombardment is involved. Overall, although wet etching is preferred for its simplicity and cost-effectiveness, dry etching is generally favored when high precision and advanced microfabrication capabilities are required.

Part B: Identify the Structure

1. Underline the **topic sentence**.
2. Write down **comparison criteria** (at least 3).
3. Highlight examples of **comparative language** (e.g., “less expensive,” “greater accuracy”).
4. Find **two examples of passive voice**.

Part C: Language Focus

Match the sentence halves:

- C 1. Wet etching is less precise because...
- a 2. Dry etching requires more complex equipment because...
- b 3. Wet etching is more cost-effective because...
- d 4. Dry etching provides higher precision because...

- a. ionized gases and plasma systems are used.
- b. simpler equipment and chemical baths are used.
- c. material is removed in all directions.
- d. anisotropic control allows finer features.

Part D: Guided Writing

Use the structure below to write your own comparison paragraph.

Sentence Frame:

- Topic Sentence:

_____ and _____ are two methods used in _____, and they can be compared in terms of _____, _____, and _____.

- Body (Comparison):

In _____, _____ is used, and it is considered _____ because _____.

In contrast, _____ is used in _____, and it is _____ because _____.

In terms of _____, _____ is _____, whereas _____ is _____.

- Conclusion:

Overall, although _____, _____ is preferred when _____.